

Part 1

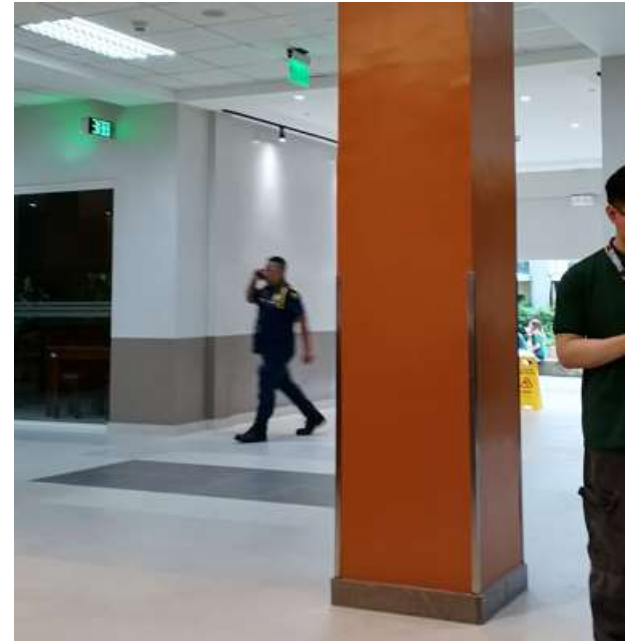
APM1110

Formative Assessment 7

Group 8

Short Summary

- The group chose to observe the number of persons who passed through the entrance of the Nicanor Reyes Hall (NRH) fronting the garden. This is located directly behind the FEU Grandstand.
- This observation is timely because the ground floor of the NRH had been recently reopened and the group would like to see the flow of people inside the building.
- Data collection was done by choosing an observation point near the entrance.
- The data collection was initially planned for 30 minutes only, but it was extended by five minutes to wait for a few more persons to pass through.



Key Parameters

Total time

= time of last person's entry — time of first person's entry
= 36 minutes, 47 seconds (Round off to 37 minutes)

Total number of persons

= 40 persons (Persons who arrived together are counted separately)

Average rate of event occurrence per unit of time

$\lambda = 40 \text{ persons} / 37 \text{ minutes}$

$\lambda = 1.0811 \text{ event occurrences per minute}$

Mean (Expected value)

$$E(X) = 1 / \lambda$$

$$E(X) = 1 / (40 / 37)$$

$$E(X) = 0.925$$

Probability density function

$$f(x) = \lambda e^{-\lambda x} \quad x \geq 0$$

Cumulative density function

$$P(X \leq x) = 1 - e^{-\lambda x}$$

Interpretations

- Using the given data, the number of persons who passed through the entrance of the NRH fronting the garden is slightly larger than 1 person per minute.
- Since this entrance sits directly behind the FEU Grandstand, this entrance was noticeably less frequented than the other ground floor entrances of the NRH.
- There could be a need to encourage people to visit the garden so that they can naturally be invited to enter or exit the NRH using the nearby entrance.